# SQL :TARGET CASE STUDY

**INITIAL EXPLORATION**

1.

```
DESCRIBE customers;

DESCRIBE orders;

DESCRIBE order_items;

DESCRIBE payments;

DESCRIBE order_reviews;

DESCRIBE products;

DESCRIBE sellers;

DESCRIBE geolocation;
```

2.

```
DESCRIBE customers;
```

Check under the type in result grid,all columns are of datatype text.

OR

```
select

   column_name,

   data_type

from information_schema.columns

where table_schema = 'target_case_study'

  and table_name = 'customers';
```

3.

```
select

   min(order_purchase_timestamp) as 'earliest_order',

   max(order_purchase_timestamp) as 'latest_order'
```

from orders;

4.

```
select
    count(distinct c.customer_city) as city,
    count(distinct c.customer_state) as state
from customers c
join orders o
    on c.customer_id = o.customer_id;
```

**IN-DEPTH EXPLORATION**

1.

```
select count(order_id) as No_of_orders,
extract(year from order_purchase_timestamp) as per_year
from orders
group by per_year
order by per_year;
```

2.

```
select
    count(order_id) as no_of_orders,
    extract(year from order_purchase_timestamp) as per_year,
    extract(month from order_purchase_timestamp) as per_month
from orders
group by per_year,per_month
order by per_year,per_month;
```

3.

```
select count(order_id) as Total_Orders,
```

```
case when extract(hour from order_purchase_timestamp) between 0 and 5 then 'Dawn'
when extract(hour from order_purchase_timestamp) between 6 and 11 then 'Morning'
when extract(hour from order_purchase_timestamp) between 12 and 16 then 'Afternoon'
else 'Night'
end as 'Time_Of_Day'
from orders
group by Time_Of_Day
order by Total_Orders desc;
```

**EVOLUTION OF E-COMMERCE ORDERS IN BRAZIL**

1.

```
select c.customer_state, count(o.order_id) as Total_Orders,
year(o.order_purchase_timestamp) as Per_Year,
month(o.order_purchase_timestamp) as Month_on_Month
from customers c
join orders o on c.customer_id=o.customer_id
group by c.customer_state,Per_Year,Month_on_Month
order by c.customer_state,Per_Year,Month_on_Month;
```

2.

```
select count(distinct customer_id) as No_Of_Customers,
customer_state
from customers
group by customer_state
order by customer_state,No_Of_Customers;
```

**IMPACT ON ECONOMY**

1.

```
select c.customer_state,year(o.order_purchase_timestamp) as Each_Year,

round(sum(oi.price),2) as total_price,

round(avg(oi.price),2) as avg_price,

round(sum(oi.freight_value),2) as total_freight,

round(avg(oi.freight_value),2) as avg_freight

from  order_items oi

join

orders o on o.order_id=oi.order_id

join

customers c on o.customer_id=c.customer_id

group by c.customer_state,Each_Year

order by  Each_Year asc,total_price desc,total_freight desc;
```

2.

```
select 2017_Cost,2018_Cost,

round(((2018_Cost-2017_Cost)/2017_Cost)*100,2) as Percentage_Increase

from (select round(sum(case

when month(o.order_purchase_timestamp) between 1 and 8 and year(o.order_purchase_timestamp)=2017 then oi.price+oi.freight_value

else 0 end),2) as '2017_Cost',

round(sum(case

when month(o.order_purchase_timestamp) between 1 and 8 and year(o.order_purchase_timestamp)=2018 then oi.price+oi.freight_value

else 0 end),2) as '2018_Cost'

from order_items oi
```

join orders o on o.order_id=oi.order_id) t;

3.

select c.customer_state,round(sum(oi.price),2) as Total_Price,

round(sum(oi.freight_value),2) as Total_Freight,

round(avg(oi.price),2) as Average_Price,

round(avg(oi.freight_value),2) as Average_Freight_Value

from order_items oi

join orders o on o.order_id=oi.order_id

join customers c on c.customer_id=o.customer_id

group by c.customer_state

order by Total_Price desc;

**ANALYSIS ON SALES, FREIGHT, AND DELIVERY TIME**

1.

select order_id,datediff(order_delivered_customer_date,date(order_purchase_timestamp)) as Delivery_Time,

datediff(order_delivered_customer_date,order_estimated_delivery_date) as Difference,

case when datediff(order_delivered_customer_date,order_estimated_delivery_date)<0 then 'Early'

when datediff(order_delivered_customer_date,order_estimated_delivery_date)>0 then 'Late'

when datediff(order_delivered_customer_date,order_estimated_delivery_date)=0 then 'On_Time'

else 'Invalid' end as Delivery_Status

from orders;

2.

Highest:

select c.customer_state,avg(oi.freight_value) as Avg_Freight

from order_items oi

join orders o on o.order_id=oi.order_id

join customers c on c.customer_id=o.customer_id

group by c.customer_state

order by Avg_Freight desc

limit 5;

Lowest:

select c.customer_state,avg(oi.freight_value) as Avg_Freight

from order_items oi

join orders o on o.order_id=oi.order_id

join customers c on c.customer_id=o.customer_id

group by c.customer_state

order by Avg_Freight asc

limit 5;

3.

Highest:

select c.customer_state,

avg(datediff(date(o.order_delivered_customer_date),date(o.order_purchase_timestamp))) as Avg_Delivery_Time

from customers c

join orders o on o.customer_id=c.customer_id

group by c.customer_state

order by avg(datediff(date(o.order_delivered_customer_date),date(o.order_purchase_timestamp))) desc

limit 5;

Lowest:

```
select c.customer_state,

avg(datediff(date(o.order_delivered_customer_date),date(o.order_purchase_timestamp))) as Avg_Delivery_Time

from customers c

join orders o on o.customer_id=c.customer_id

group by c.customer_state

order by avg(datediff(date(o.order_delivered_customer_date),date(o.order_purchase_timestamp))) asc

limit 5;
```

4.

```
select c.customer_state,round(avg(datediff(date(o.order_delivered_customer_date),date(o.order_estimated_delivery_date))),0) as Avg_Delivery_Time

from customers c

join orders o on o.customer_id=c.customer_id

group by c.customer_state

having Avg_Delivery_Time <0

order by Avg_Delivery_Time asc

limit 5;
```

**ANALYSIS BASED ON PAYMENTS**

1.

```
select count(distinct o.order_id) as No_of_Orders,year(o.order_purchase_timestamp) as Per_Year,

month(o.order_purchase_timestamp) as Per_Month,p.payment_type

from payments p

join orders o on o.order_id=p.order_id
```

```
group by Per_Year,Per_Month,p.payment_type

order by Per_Year,Per_Month,No_of_Orders desc;
```

2.

```
select count(distinct o.order_id) as No_of_Orders,

p.payment_installments

from payments p

join orders o on o.order_id=p.order_id

group by p.payment_installments

order by p.payment_installments desc;
```

## INSIGHTS AND RECOMMENDATIONS

Insight 1:

Orders **increased strongly from 2016 through 2018**, but **declined sharply** toward the end of the available period. Orders fell by approximately 99.75% from August to September 2018, followed by a further 75% decline from September to October..

Recommendations:

The business should investigate the reasons behind this sharp decline by examining **seasonal factors, delivery performance, and order trends across its major states** during September and October. Since these are the final months in the available dataset, data completeness should also be considered before concluding that the decline reflects an actual drop in business activity.

Insight 2:

Customer ordering activity is highest during the **6 PM–11:59 PM** period accounting for approximately 41% of all orders . During these hours, the highest-order-volume categories include **bed table bath, HEALTH BEAUTY, sport leisure, Furniture Decoration, and computer accessories**, while categories such as **food, fashion, and men's clothing** have comparatively lower order volumes.

Recommendation:

The company could use these peak ordering period to **promote lower-order-volume product categories** such as food, fashion, and men's clothing, with the aim of increasing their visibility and order volume.

Insight 3:

SP, RJ, and MG together contribute **63.37% of total revenue**, with **SP alone accounting for 38.28%.** In comparison, RS (5.52%), PR (5.03%), and SC (3.83%) contribute smaller individual shares but **together account for 14.38%** of total revenue, indicating potential for further growth in these markets

Recommendation:

For established high-performing markets such as SP, RJ and MG, focus on retaining the existing customer base by maintaining efficient delivery and product availability. For **growing markets such as RS, PR, SC and BA**, **increase targeted customer promotions** to capitalize on their growth potential.

Insight 4:

RR, PB, RO, and AC have the **highest average freight costs**, and these states also have relatively **high average actual delivery times**.

Recommendation:

Target should investigate logistics in these states by looking at **shipping routes, delivery partners, fulfillment locations, and freight costs** to identify opportunities to reduce both delivery time and shipping expenses. It could also work with local delivery partners that may offer faster delivery at lower costs.

Insight 5:

Customers predominantly prefer lower installment plans, with **84.69% of orders using five or fewer installments**. In contrast, orders using higher installment counts are significantly less common, **less than 0.1% of orders**.

Recommendation:

Target can analyse the products associated with higher installment counts and work with financing partners to offer more economical installment plans. This could make higher-value purchases more affordable and potentially encourage more customers to purchase.

Insight 6:

**14.20%** of reviewed orders received a **rating below 3 stars** and they seem to have a direct relationship with **longer delivery times**.

Recommendation:

The company should **investigate further** on the causes of lower user reviews and **provide actionable insights** to improve the rating with **particular attention to delivery performances.**